Directive System for Maps' Guided System Based on API

Criterion A - Planning

Defining the problem: [Refer to **Appendix I – Initial Interviews**]

My client, Mr. William is a high school physics teacher who everyday drive a car to school and back to home. As he concerns, every day when he drives a car to school is very inconvenient because each day, the traffic condition is different, so when he continue his formal daily routine, sometimes there will be traffic jam and greatly extend the time he reaches school. Also, if he wants to purchase for breakfast, he does not know if there's any breakfast store. Which means he may waste a lot of time during the searching process of his breakfast, so he is unable to reach the school on time. Some maps' guided system is very inefficient that sometimes the route maybe chooses automatically the longest because the cost can be cheapest.

In order to solve his problem, I am going to design a system to help him verify the daily routine and the best-fit route for his special cases. For instance, when he wants to have breakfast and purchase newspaper in the early morning, the system will automatically recommend the route with breakfast store and book shops while the distance is the shortest to reach his destination.

Rationale for proposed solution:

I am eager to make my client life much easier and to complete this online website with JavaScript by html file. JavaScript is the best choice because it's wider and more convenient to use the ability to construct a website and change the font of the word size. Beside JavaScript, Amap (Chinese local map service for navigation) API (Application Programming Interface) can be used as a part of my program because they their data of mapping can be more accurate, so in order to provide a more preferable service to my clients. As well as JavaScript, Nodejs allows JavaScript to not only run on webpage, but also on server. Since Nodejs is single thread, suitable for "I/O(input or output)"[1] intensive applications, only occupy small internal storage.

To design a GUI is also my target. This is because each user should have their own account, so that their information of research history won't be mess-up together. Other than have a place to preserve the users' information to provide convenience to people, and also being helpful for my clients' family members and students.

JavaScript as my main programming language compare with Java, python and C++. This is because JavaScript is more functional when forming a html file. As it is compatible for all browsers, enable users to insert different API into the website and make more usages of it.

Wordcount: 520

Success Criterions:

The intended application should:

1. Have the ability to login, preserve and protect the data.
2. Construct a user-friendly GUI.
3. With a necessary text-field to enter searching information.
4. The users can clearly check their research and travelling history, and edit.
5. Able to present different types of travelling (bus, car, feet).
6. Enable users to check their own point of interest (POI).
7. Able to check the surroundings of destination and starting point.
8. System is able to locate users' current position.
9. Users are able to choose different modes.
10. Accept multiple-users' access.

[1] I/O-intensive applications are a critical consideration in backend computing. They revolve around tasks and processes that heavily rely on input/output